

Power vs. Planet: Examining HPC's Environmental Impact

High-Performance Computing (HPC) systems play a pivotal role in advancing scientific research, artificial intelligence, and big data analytics. However, their substantial energy consumption and environmental impact have become significant concerns [1]. This study presents a comprehensive model that quantifies the environmental impact of HPC systems in the present and the future, focusing on **energy consumption, carbon emissions, and water usage**, providing suggestions for reducing the environmental impact of HPC systems.

Our first model for energy consumption of HPC systems begins at a **component level**, including processors, memory, storage, and networking equipment. Utilizing the formula from O'Brien et al. [2], we accounted for **different power levels** (idle and max) and **utilization rates** to reflect real-world operating conditions. Furthermore, we incorporated **Power Usage Effectiveness (PUE)** metric since we recognize that HPC facilities consume energy beyond IT components. Our result is further validated with empirical data on data center energy consumption proportions. The total energy consumption of HPC systems in 2024 was estimated to be 390.411 TWh, accounting for approximately **1.304% of global electricity consumption**.

Building upon the energy consumption results from our first model, we conducted a comprehensive Life-Cycle Assessment (LCA) of carbon emissions for HPC systems, which includes three key stages: **production, operation, and disassembly**. Operational emissions were calculated by linking energy consumption with the **carbon intensity** of various energy sources. Production emissions were estimated using carbon footprint data for **integrated circuits**, while disassembly emissions were determined based on the annual disposal (burning) of processors. Our calculation indicates that in 2024, carbon emissions due to the HPC system are about 187.7 million tons of CO₂, which represents about **1.31% of worldwide carbon emissions**. Our sensitivity analysis suggests that HPC carbon emissions are **highly responsive** to changes in renewable energy percentage in electricity generation, and a transition to a higher renewable energy mix can significantly reduce emissions.

A simple snapshot of current carbon emissions is insufficient to capture the long-term environmental effects of HPC systems, given evolving factors like energy efficiency, increasing power efficiency, increasing demand and shifts in renewable energy usage. To address this, we refined our model to account for changes in PUE, processors' power, market expansion, and carbon intensity. Using the **Grey Prediction Model**, we analyzed PUE trends despite limited data, while a **modified Gaussian curve** projected carbon intensity through 2030. We placed importance on GPU upgrades: while newer GPUs deliver energy efficiency, their increased power demands can paradoxically drive higher energy consumption. Our model predicts a significant **linear decline** in total carbon emissions—dropping to **116 million tons** by 2030—largely driven by the efficiency gains associated with GPU upgrades. Sensitivity analysis revealed a slight sensitivity to processor power proportions, confirming the robustness of our results.

Beyond carbon emissions, we examined **water consumption** as another critical environmental factor, focusing on two primary sources: electricity supply and cooling systems. They were analyzed using the United States as a case study, with water consumption of cooling system categorized into three methods: air cooling, direct liquid cooling, and phase change cooling. By avoiding double counting and leveraging industry data, we calculated that water consumption due to electrical supply to be 32920 liters, while annual water consumption at 507.06 billion liters for air cooling, 81.83 billion liters for direct liquid cooling, and 37.56 billion liters for phase change cooling. Our approach highlights the **significant water demands of HPC systems** and emphasizes the need for more water-efficient cooling technologies.

Keywords: HPC; Carbon Footprint; Energy Consumption; Carbon Emission; Water Consumption; Grey Prediction Model

INCLUSION of HPC ENVIRONMENTAL

IMPACTS in 2030 GOALS

Team #15429

Honored Members of the United Nations Advisory Board,

Our team is writing to express a critical concern regarding the recent report titled "Governing AI for Humanity" released in September 2024. While the report comprehensively addresses various facets of artificial intelligence (AI), it omits the discussion on High-Performance Computing (HPC) systems and their environmental impacts. Given the pivotal role HPC plays in advancing AI and other computational sciences, our team urges the Board to include a detailed section on the environmental ramifications of HPC in the upcoming developmental goals for 2030.

Through a comprehensive life-cycle assessment of HPC systems, we have uncovered significant insights into their global energy demands and carbon footprint.

1. HPC systems consume approximately 390 billion kWh in 2024, accounting for about 1.3% of global electricity usage.

2. The combined emissions from the operation, production, and disassembly of HPC systems amount to approximately 187.7 million tons of CO₂ in 2024, representing 1.31% of worldwide carbon emissions.

3. The growing demand for HPC in areas like AI is expected to significantly increase energy consumption in the future, while actions such as increasing the proportion of renewable energy in electricity generation and upgrading GPUs to more energy-efficient models can substantially reduce carbon emissions.

4. Beyond carbon emissions, HPC systems require significant amounts of water for cooling, especially in regions like the United States.

We deem the environmental impact of HPC systems as a global sustainability concern that intersects with the United Nations' Sustainable Development Goals (SDGs), particularly those related to climate action (SDG 13), affordable and clean energy (SDG 7), and responsible consumption and production (SDG 12). Therefore, we propose the following actionable steps to mitigate the environmental impact of HPC systems.

1. Set strict WUE (Water Usage Effectiveness) standards and implement fines for non-compliance.

2. Encourage investment in the development and adoption of more energy-efficient GPUs.

3. Advocate for policies that substantially increase the global share of renewable energy in the electricity mix. Ensure a balanced approach to maintain sufficient electricity generation to meet growing HPC demands.

4. Support research and implementation of advanced cooling technologies that reduce both energy and water consumption.

Sincerely,

Team #15429

Contents

1	Introduction and Background	3	6.1.3	HPC Market Growth and Increase in IT components	14
1.1	Background	3	6.1.4	Predicting Worldwide Carbon Intensity	14
1.2	Problem Restatement	3	6.1.5	Predicting Total Carbon Emission Operational Stage	15
2	Assumptions and Justifications	3	6.2	GPU Upgrades and Increased Disassembly	16
3	Key Terms and Symbols	4	6.3	Results from 2025 to 2030	18
4	Model for HPC Energy Consumptions	4	6.4	Sensitivity analysis	18
4.1	Energy Formula	4	7	Model of Impacts of Water Usage	19
4.2	Using PUE Metric	5	7.1	Why This Aspect?	19
4.3	Breaking Down E_{IT} to Find Energy Consumption	5	7.2	Water Consumption of Electricity Generation	19
4.4	Results and Relevance	6	7.3	Water Consumption of HPC Systems	20
5	Model of HPC Carbon Emissions	6	7.3.1	Total Water Consumption	21
5.1	Model Structure Overview	7	7.4	Water Pollution of HPC Systems	21
5.2	Carbon Emission due to Production	7	7.5	Water Consumption and Pollution of HPC Systems in the USA	22
5.3	Carbon Emission due to Operation	8	7.6	How It Relates to Other Aspects?	22
5.4	Carbon Emission due to Disassembly	10	8	Recommendations to Reduce Environmental Impact of HPC	22
5.5	Results and Analysis	10	8.1	Technical Solutions	22
5.6	Sensitivity Analysis of Energy Mixes	11	8.2	Policy-Oriented Solutions	22
5.7	Effects and Challenges of Switching to 100% Renewable Energy	11	8.3	Incorporate the Recommendation to Our Model	23
6	Model of Estimating Future Trend	12	9	Strengths and Weaknesses	23
6.1	Predicting Carbon Emission of Operational Stage	12	9.1	Strengths	23
6.1.1	Trend in PUE	12	9.2	Weaknesses	23
6.1.2	Predicting Processor Power Growth	13	10	Discussion	23
			11	<u>Report on Use of AI</u>	25

1 Introduction and Background

1.1 Background

As computational demands continue to grow worldwide, high-performance computing (HPC) has emerged, which links numerous computers together to achieve large-scale computational capabilities. In recent years, the rapid development of artificial intelligence and big data has further increased the demand for HPC, with the market expanding by about \$1.1 billion from 2020 to 2021.

However, this surge in HPC usage carries significant environmental implications, particularly concerning **energy consumption, carbon emissions, and water usage**. The computational and cooling requirements of HPC systems necessitate vast amounts of electricity, often sourced from non-renewable energy grids, which contributes directly to global CO₂ emissions. Moreover, the cooling processes employed in HPC facilities often involve substantial water consumption, presenting a dual environmental challenge. These systems not only place additional pressure on local water resources but also risk contaminating water supplies through the release of thermal or chemical effluents.

Given the United Nations Sustainable Development Goals (SDGs) by 2030, it is crucial to address the environmental impact of HPC systems. Therefore, this paper aims to explore the environmental impacts of HPC computing in recent years, analyze trends in energy and resource consumption, and propose sustainable practices to align HPC development with global environmental objectives.

1.2 Problem Restatement

Our model seeks to quantify the current and future environmental impact of HPC systems, focusing on energy consumption and its effects. We interpret the problem as the following **four** fundamental questions:

Questions.

- (1) What are the key components contributing to the energy consumption of High-Performance Computing (HPC) systems, and what is the total annual energy consumption (2024) of HPC worldwide?
- (2) What are the total carbon emissions associated with the energy consumption of HPC systems, taking into account the full life cycle of these systems?
- (3) How will carbon emissions evolve in the future, considering the growing demand for HPC, advancements in HPC technology, and the optimization of energy mixes?
- (4) Beyond carbon emissions, what is another environmental impact of HPC systems that are influenced by energy consumption?

2 Assumptions and Justifications

Assumption 1: We assume that HPC systems operate continuously, running 24 hours a day, 365 days a year, regardless of variations in operational power.

Justification 1: HPC systems are typically used for high-demand computations and are shown to operate continuously with minimal downtime[4]. Additionally, restarting HPC systems often requires significant and unpredictable energy consumption, leading to non-negligible costs[5]. However, there are still differences in power consumption during periods of idling and peak performance.

Assumption 2: We assume that all renewable energy sources have negligible carbon emissions, and therefore, renewable energy does not contribute to carbon intensity.

Justification 2: Non-renewable energy sources—such as coal, oil, and natural gas—are the primary contributors to carbon emissions. Although renewable energy involves some emissions, they are negligible compared to non-renewable energy[6].

Assumption 3: We assume that all electricity used by HPC systems comes from a globally unified electricity pool, where electricity generated in all regions is aggregated and then distributed based on demand.

Justification 3: Due to limited data on the specific locations and regional power distribution of HPC sys-

tems, it is not feasible to account for the differences in electricity generation or carbon intensity across regions. Therefore, we consider global electricity as a single aggregated source and calculate its carbon intensity based on the global average. While this assumption simplifies the modeling process, it may obscure regional disparities, making the results more applicable to some regions than others.

Assumption 4: We assume that all components (e.g., GPUs and CPUs) perform consistently in accordance with the standard performance metrics provided by the manufacturers during operation.

Justification 4: In real-world conditions, components such as GPUs and CPUs may experience performance variations due to environmental factors, workload changes, or aging hardware. Nonetheless, we base our modeling on the assumption that all components operate as specified by their manufacturers. Although this assumption may introduce minor inaccuracies, it ensures the stability and reliability of our calculations.

Assumption 5: We assume that all globally relevant data are equivalent to the data retrieved from our selected database.

Justification 5: Datasets like the *Top 500* and *Green 500* only include a small number of top-tier HPC systems, which are not representative of most HPC systems. Instead, we use databases that focus on global energy production, supply, and consumption data, reflecting the energy usage of the majority of HPC systems. However, this assumption may exclude atypical systems and underestimate their impact.

3 Key Terms and Symbols

Symbols	Definition	Units
C	Produced Carbon / Carbon Emission	t
CI	Carbon Intensity	gCO_2/kWh
E	Energy	kWh
EG	Energy Generation	kWh
K	Capacity of HPC Systems' Components	MB
N	Number	NA
P	Power	kW
PUE	Power Usage Effectiveness	NA
$R_{C:X}$	Ratio of Fuel Type X's Carbon Emission to Total Carbon Emission	%
$R_{EG:X}$	Ratio of Country X's EG to Worldwide EG	%
T	Temperature	K
t	Time	h
W	Water Consumption	L
WR	Water Consumption Rate	L/kWh

Table 3.1: Symbols and Variables

4 Model for HPC Energy Consumptions

HPC systems are characterized by their significant energy demands. In this context, energy is defined as the **amount of electricity consumed during the operation of HPC systems**. HPCs electrical consumption is a **major contributor to carbon emissions**, as electricity generation often relies on fossil fuels. To accurately assess the carbon emissions associated with HPC operations, it is crucial to estimate their annual electricity consumption. This model aims to estimate the **electricity consumption of HPC systems worldwide for 2024**.

4.1 Energy Formula

The inspiration for energy consumption calculation comes from the principle of physics:

$$E = P \times t \quad (4.1)$$

where E is the energy consumed, P is the power consumption, and t is the time of operation. This simple principle of multiplying power by time forms the basis of how we calculate the energy consumption for individual compo-

nents in an HPC system. IT Components such as CPUs and GPUs consume power proportional to their utilization. Following *Assumption 1*, which assumes HPC systems operate continuously, we calculate energy consumption over an entire year for consistency and accuracy.

4.2 Using PUE Metric

Calculating HPC power consumption requires more than summing up IT components' energy use; non-IT infrastructure like cooling and lighting also consumes significant energy. Cooling alone accounts for about 50% of data center energy use [7]. To capture the total facility energy consumption, including IT and non-IT components, we use Power Usage Effectiveness (PUE), which measures efficiency by comparing total facility energy use to IT equipment energy consumption:

$$\text{PUE} = \frac{E_{\text{facility}}}{E_{\text{IT}}} \quad (4.2)$$

where E_{facility} is the total energy used by the data center (including IT equipment, cooling, and infrastructure), and E_{IT} is the energy used solely by IT equipment like servers and networking. A PUE value of 1.0 indicates ideal efficiency, meaning all the energy consumed by the facility is used solely by the IT equipment with no overhead from other infrastructure. In reality, PUE values are always greater than 1.0. Re-expressing Equation 4.2, we obtain the total consumption of the facility, which is consistent with the energy consumption method in [8].

$$E_{\text{facility}} = \text{PUE} \times E_{\text{IT}} \quad (4.3)$$

4.3 Breaking Down E_{IT} to Find Energy Consumption

Following the energy predictive model developed by Kenneth O'Brien et al.[2], we divide the **IT equipment energy consumption** (E_{IT}) into four core components: **processors** (including CPUs and GPUs), **memory**, **disk**, and **network**. These components represent the primary sources of power consumption, enabling computational tasks within HPC systems. To meet the energy demand of these systems, a large amount of electricity is produced, which in turn leads to the generation of carbon dioxide. Figure 4.1 illustrates the breakdown of E_{IT} into its various components.

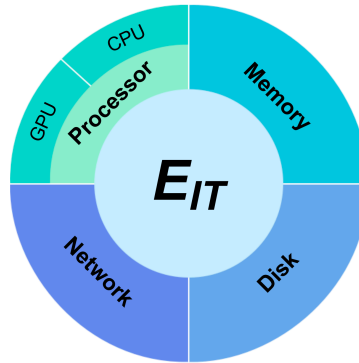


Figure 4.1: Components of E_{IT}

While HPC systems are continuously powered on, as in Assumption 1, they do not always operate at full computational load, or **full capacity**. In fact, studies have shown **significant differences between operational power in different workloads**. For instance, the power consumption of blade processors in high-demand scenarios like SPECfp reaches more than twice the level observed in the idle state, which illustrates a substantial idle-to-peak (benchmark) difference [9].

Therefore, to reflect real-world fluctuations in component power, we use the calculation method in [2] to capture the difference in power and utilization:

$$P_{\text{IT},i} = (P_{i,\text{idle}} + (P_{i,\text{max}} - P_{i,\text{idle}}) \times U_i) , i \in \{\text{CPU, GPU, Memory, Disk, Network}\} \quad (4.4)$$

where $P_{i,\text{idle}}$ is the idle (base) power consumption of component i . $P_{i,\text{max}}$ is the maximum power consumption of component i , and U_i represents the utilization rate of component i , a value between 0 and 1. This equation accounts for the difference between idle and maximum power consumption by multiplying **the additional power**

$(P_{i,\max} - P_{i,\text{idle}})$ with the component's **utilization rate**. This quantifies the **extra power consumed when the component transitions from an idle state to full utilization**. The resulting power is between minimum power ($P_{i,\text{idle}}$) and maximum power ($P_{i,\max}$), which provides a realistic bound:

$$E_{\text{component}} = (P_{i,\text{idle}} + (P_{i,\max} - P_{i,\text{idle}}) \times U) \times t \quad (4.5)$$

where t is the time period over which the component operates, which is held constant to all the hours in a year by Assumption 1. Therefore, total energy consumption is obtained by adding the consumption of each component, illustrated in Equation 4.6.

$$E_{\text{IT}} = \sum_{i \in \text{IT components}} (\text{Energy Consumption of } i) \quad (4.6)$$

Expanding this equation, we get the total energy of E_{facility} :

$$E_{\text{facility}} = \text{PUE} \times \sum_{i \in \text{IT components}} ((P_{i,\text{idle}} + (P_{i,\max} - P_{i,\text{idle}}) \times U_i) \times t \times N_i) \quad (4.7)$$

where $IT_{\text{components}}$ refers to CPU, GPU, memory, disk, and network.

4.4 Results and Relevance

To estimate the maximum power of a CPU or GPU, we use the **Thermal Design Power (TDP)**, which represents the peak thermal load during heavy workloads. Combining idle power, maximum power, and utilization rates of common CPUs and GPUs, we calculate average power. Annual energy consumption is then derived and scaled by the number of processors to obtain total processor energy consumption. Since processors constitute 30% of total IT energy usage [7], we estimate IT energy consumption proportionally. Relevant data is available in our database.¹²

The facility's total energy consumption, E_{facility} , is calculated by multiplying IT energy consumption by the 2024 Power Usage Effectiveness (PUE) of 1.56 [3]. This yields a total HPC energy consumption of **390.411 TWh**, a significant share of global electricity consumption (**29,924,700 TWh** in 2024 [10]). The proportion of global electricity used by HPC systems is shown in Equation 4.8:

$$\rho_{\text{HPC}} = \frac{E_{\text{facility}}}{E_{\text{world}}} \quad (4.8)$$

where ρ_{HPC} is the share of HPC in global energy usage, and E_{world} is global electricity consumption. Our calculation gives $\rho_{\text{HPC}} = \mathbf{1.304\%}$, aligning with empirical data, such as the 2022 report estimating data centers' consumption at **2%** of global electricity [21]. This demonstrates the reliability of our estimate for HPC systems' global energy share.

5 Model of HPC Carbon Emissions

Having established the energy consumption of HPC systems, we now transition to analyzing the carbon emissions using a **Life Cycle Assessment (LCA)** framework. LCA is a comprehensive methodology for evaluating the environmental impact of a product, process, or service throughout its **entire lifecycle**. By considering three stages—from material acquisition and manufacturing to operation and disposal—LCA provides a **holistic view** of carbon emissions. This can serve as a critical tool for businesses and policymakers to inform decisions.

Drawing from Baolin Li's study of HPC carbon emission, we use LCA to focus on the carbon emissions generated by HPC systems over one year. We analyze the carbon footprint from two primary perspectives: **embodied carbon footprint** and **operational carbon footprint**[10]. Embodied carbon refers to emissions resulting from "one-time activities" mostly the initial stages and final stages of HPC systems' lifecycle. We further characterize the embodied carbon footprint into **manufacturing emissions** and **disassembly emissions**. Operational emissions come from the electricity consumption during system operation, which is related to energy consumption. These emissions arise during the process of electricity generation, which is required to support HPC operations.

¹https://github.com/reformship/hpc_research/blob/main/CPU.xlsx

²https://github.com/reformship/hpc_research/blob/main/GPU.xlsx

5.1 Model Structure Overview

Since **energy consumption** is the dominant factor driving carbon emissions, our model builds on the results of our previous energy consumption analysis and incorporates the varying carbon intensities of energy sources. Following our categorization of carbon emission, the overall model is defined as follows:

$$C_{\text{total}} = C_{\text{prod}} + C_{\text{op}} + C_{\text{dis}} \quad (5.1)$$

where C_{total} represent the total carbon emissions from HPC, C_{prod} represent the carbon emissions from production, C_{op} represent the operational carbon emissions, and C_{dis} represent the carbon emissions from disassembly. Figure 5.1 visually illustrates the key concepts and relationships within our model.

While C_{prod} and C_{dis} directly contribute to carbon emissions, C_{op} contributes indirectly: carbon is primarily generated from the **electricity** required to continually power HPC systems. To calculate C_{op} , we convert energy consumption to carbon emissions based on the **carbon intensity** of electricity generation. These carbon intensity values are categorized by country and energy source, capturing variations in national energy profiles and diverse generation methods.

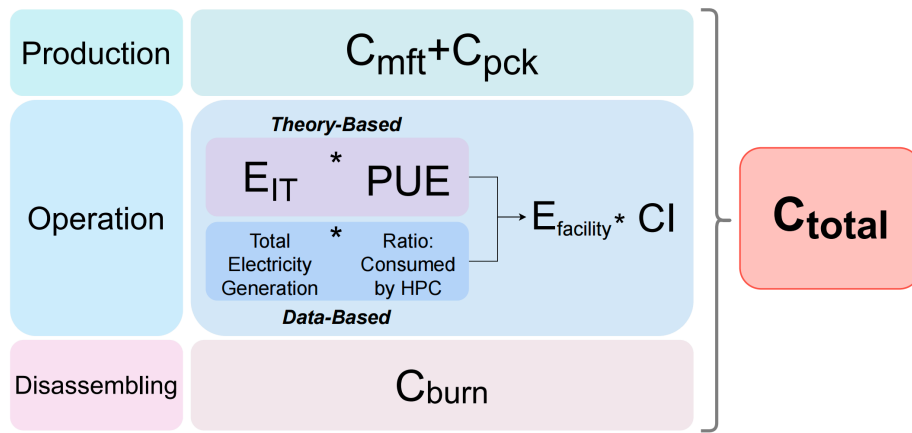


Figure 5.1: Life Cycle Assessment of HPC Carbon Emissions

5.2 Carbon Emission due to Production

High-Performance Computing (HPC) systems, from design to deployment, undergo multiple stages of manufacturing and assembly, each contributing to CO₂ emissions. In order to calculate carbon emission due to production more conveniently, we split it into **storage** and **processor** emissions. Therefore, total carbon emissions from production (C_{prod}) are expressed as:

$$C_{\text{prod}} = C_{\text{processor}} + C_{\text{storage}} \quad (5.2)$$

Building upon [10], we further categorize processor-related emissions into two primary components: **manufacturing** and **packaging**.

The carbon emissions from **processor packaging** are given by Equation 5.3, according to previous studies on the carbon footprint of HPC systems across their lifespans [4]:

$$C_{\text{pck}} = C_{\text{IC}} \times N_{\text{IC}} \quad (5.3)$$

where N_{IC} is the number of integrated circuits produced for CPUs and GPUs, and C_{IC} is the carbon emission per integrated circuit. C_{IC} is 150 about gCO₂ per unit. To determine the **total carbon emissions from processors**, we consider the proportion of packaging emissions relative to the total production emissions. This is given by:

$$C_{\text{processor}} = \frac{C_{\text{pck}}}{P_{\text{ac}}} \quad (5.4)$$

where P_{ac} here stands for the percentage of packaging carbon in the total carbon of production, which is 12%[4]. For **storage systems**, total emissions due to manufacturing and packaging are calculated based on their **storage capacity**:

$$C_{\text{storage}} = C_{\text{cap}} \times K_{\text{disk}} \quad (5.5)$$

where C_{cap} stands for the emission per capacity of a certain type of storage disk, and K_{disk} stands for the capacity of the disk. After we calculate the carbon emission from both storage and processors, we get the final carbon emission of HPC production[10, 11, 12]. Combining the contributions from processors and storage systems, the total production-related carbon emissions for HPC systems can be expressed as:

$$C_{prod} = \left(\frac{C_{IC} \times N_{IC}}{P_{ac}} \right) + (C_{cap} \times K_{disk}) \quad (5.6)$$

where the left term represents carbon emissions from processor production, and the right term calculates the carbon emissions from storage system production.

5.3 Carbon Emission due to Operation

To calculate the operational carbon emissions, we build upon the results from our previous energy consumption model. By multiplying the total energy consumption of HPC systems by the **carbon intensity** of electricity generation, we convert energy usage into carbon emissions.

Electricity generation is a known source of carbon emissions, but it is crucial to recognize that **not all energy sources involved in electricity production release carbon dioxide**. Based on Assumption 2, nonrenewable energy is identified as the main driver of carbon emissions. As illustrated in Figure 5.2, there has been an increase in renewable energy with a decline in nonrenewable energy over time. This shift indicates a potential reduction in carbon emissions from electricity consumption in the future. It is important to note that nuclear energy, although classified as **nonrenewable**, does not emit carbon dioxide during electricity generation [13]. For this reason, nuclear energy is not considered in terms of carbon emissions.

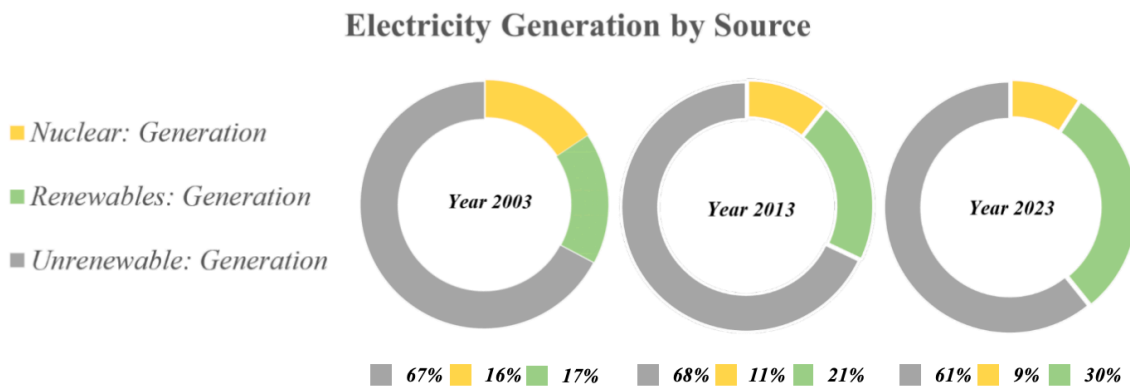


Figure 5.2: Electricity Generation by Source.³

To account for the variability in electricity generation methods, we introduce the concept of **carbon intensity of electricity (CI)**. Carbon intensity measures the *amount of carbon dioxide, in grams, released per kilowatt-hour (kWh) of electricity generated*. It reflects the emissions associated with electricity production based on the energy mix. For example, grids relying heavily on coal will have higher carbon intensity compared to those dominated by renewable energy sources.

Monitoring carbon intensity is critical for decarbonizing the energy sector and achieving global climate goals [14]. Figure 5.3 shows the trend of CI worldwide. The graph shows a sharp increase in carbon intensity from 2000 to 2010, driven by rising global electricity demand largely met by coal and natural gas. After 2015, carbon intensity began to decline due to policies supporting international climate agreements like the Paris Agreement [15]. The significant drop after 2020 reflects global efforts to transition to low-carbon energy sources. As the energy mix continues to shift, carbon intensity is expected to decline further, a trend explored in detail in the section **Effects and Challenges of Switching to 100% Renewable Energy**.

Electricity generation and carbon intensity data were collected for each country [16]. Given the substantial variations in electricity generation (EG) and carbon intensity (CI) across nations, the **weighted average method** was applied to ensure that **countries have a fair representation of their contributions to global carbon intensity**.

³Source: https://github.com/reformship/hpc_research/blob/main/EI-Stats-Review-All-Data.xlsx

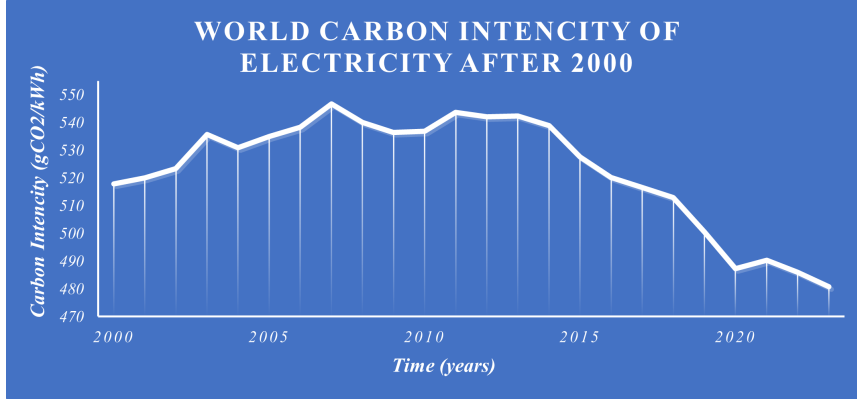


Figure 5.3: Carbon Intensity World Trend

The total global electricity generation (EG) was calculated by summing the electricity generation of all countries. Each country's contribution to global electricity generation is then represented as a proportion, shown as Equation 5.7. $R_{EG:X}$ denotes the weight of a country's electricity generation EG_X relative to the global total. The weighted average method ensures that **countries with higher electricity production have a proportionally greater influence on the world's carbon intensity**.

Additionally, the proportion of carbon emissions contributed by each fuel type to the total carbon emissions is determined. Here, the weighting reflects the share of electricity generated by each fuel type, accurately representing its contribution to overall emissions. To refine the analysis, the **proportion of carbon emissions contributed by each fuel type** was calculated. This proportion reflects the share of electricity generated by each fuel type and is expressed as Equation 5.7.

$$R_{EG:X} = \frac{EG_X}{EG_{World}} \quad R_{C:X} = \frac{C_X}{C_{total}} \quad (5.7)$$

Using this proportion ($R_{C:X}$), the average carbon intensity for a given country was calculated as:

$$CI_{Fuel} = \sum_{i=1}^{X_1} R_{C:i} \cdot CI_i \quad (5.8)$$

where X_1 is the set of all fuel types used to produce electricity, $R_{C:i}$ is the proportion of emissions from fuel type i , and CI_i is the carbon intensity of fuel type i .

Ultimately, the result is obtained by multiplying each proportion (R_{EG}) by its corresponding carbon intensity across all countries:

$$CI_{world} = \sum_{j=1}^{X_2} R_{EG:j} \times CI_j \quad (5.9)$$

Here, X_2 represents the set of all countries in the world, $R_{EG:j}$ is the proportion of electricity generated by country j , and CI_j is the carbon intensity of country j .

After determining the energy consumed by HPC systems and the carbon intensity of electricity, the total carbon emissions produced during the operational stage can be calculated according to *Assumption 3*.

$$C_{op} = E_{IT} \times CI_{world} \quad (5.10)$$

Expanding the equation 4.9, we get the expanded result for C_{op} in equation 4.10.

$$C_{op} = CI_{world} \times PUE \times \sum_i ((P_{i,idle} + (P_{i,max} - P_{i,idle}) \times U_i) \times T \times N_i) \quad (5.11)$$

where i refers to CPU, GPU, memory, disk, and network.

5.4 Carbon Emission due to Disassembly

After certain components of HPC have passed their usable ages, new components will come up, and the old ones will be recycled. However, not all electric garages would be fully recycled. According to ITU's research, only 17.4% of the world's electric waste would be recollected, and others would be discarded or burned. The uncollected would emit a huge amount of CO₂ into the atmosphere. Meanwhile, directly dumping the wastes into land does not result in any short-term carbon emissions. Therefore, we consider carbon emissions due to disassembly as equivalent to carbon emissions due to burning:

$$C_{dis} = C_{burn} \quad (5.12)$$

For the C_{burn} , we calculate it separately with respect to components. We would time the number of specific components dumped that year with the carbon emission per number of those components burned. Therefore:

$$C_{component} = C_B \times N_B$$

where C_B stands for carbon per number burned of each component and N_B means the number of each component burned.

And as a result, we can calculate the carbon emission caused by disassembly through:

$$C_{dis} = \sum_{i \in components} C_{B,i} \times N_{B,i}$$

i here stands for individual burned component types such as GPU.

5.5 Results and Analysis

To calculate C_{prod} , we summed the carbon emissions from the manufacturing and packaging of processors and storage components. Using data from the SEMI report [19], we determined that packaging emissions for integrated circuits (ICs) are **150 g CO₂ per unit**. Therefore, the total emissions from processor packaging, including CPUs and GPUs, are calculated as:

$$C_{processor} = 150 \times N_{IC} \quad (5.13)$$

This result is approximately 23,265 tons of CO₂ emissions. Multiplying the capacity of the disk with carbon emission per capacity gives us the emission of storage systems, which is 12434 tons. Therefore, the total emissions for HPC production is approximately 37,500 tons of CO₂. The breakdown is illustrated in Table 5.1.

	Processors	Storage
Carbon Emission (Tons)	23265	12434

Table 5.1: Results for C_{prod} in 2024

For the operational stage, we calculated C_{op} by multiplying the total facility energy consumption ($E_{facility}$) with the global carbon intensity (CI) of electricity. From our previous model, the annual energy consumption of HPC facilities is estimated to be **390.411 TWh**. Using the global carbon intensity of **480.84 g CO₂/kWh**, our result demonstrates that operational carbon emissions in 2024 are approximately 187.7 million tons, representing **1.31%** of global carbon emissions.

To estimate C_{dis} , we calculated emissions from the burning of HPC components, assuming a maximum 5-year lifespan for GPUs and a disposal rate that increases with age. It is established that burning one CPU or GPU generates 250 g CO₂ [10]. Using collected data on component lifespans and disposal probabilities, we calculated the total emissions from disassembly. The result is illustrated in Table 5.2.

Summing the carbon emissions together, the 2024 carbon emissions due to HPC system are estimated to be **187.74 million tons**. The results reveal several important insights.

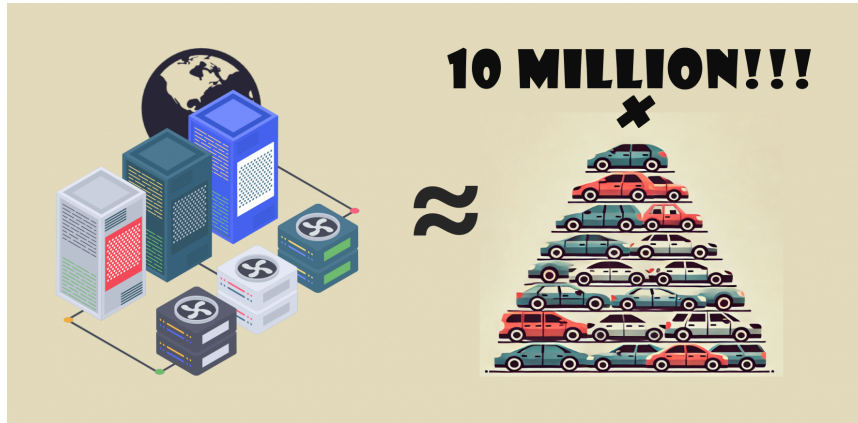
Operational emissions dominate the total carbon footprint, accounting for nearly 99% of the overall emissions. This outcome is logical, as the operation of HPC systems occurs continuously throughout the year, leading

Burned HPC Components	CPU	GPU	Storage	Total
Carbon Emission (tons)	12	600	1.59	613.59

Table 5.2: Carbon Emission Produced by Each Type of Component

to significantly higher emissions compared to the one-time contributions from production and disassembly. This underscores the importance of **improving energy efficiency** and **reducing carbon intensity in electricity generation** to mitigate the environmental impact of HPC systems in the future.

To contextualize the carbon emissions produced by HPC systems, the emissions from these systems can be equated to those generated by approximately ten million passenger vehicles.



5.6 Sensitivity Analysis of Energy Mixes

In this section, we will focus on how much change in energy mixes affects our HPC carbon emission model. The **shift towards renewable energy sources** has become a central component of global efforts to mitigate climate change and reduce carbon emissions. We are interested in the impacts of switching the **percentage of renewable energy share in electricity generation** on the carbon emission of HPC and analyze the sensitivity of our model to energy mixes.

We record the carbon emission calculated by our model under current conditions. Then, we add 1% renewable energy in electricity generation and repeat to record the change in output of our model until the percentage of renewable energy becomes 100%. Figure 5.4 shows the reduction of carbon emissions due to the changing percentage of renewable energy, and Figure 5.5 shows the trend in carbon emissions as the renewable energy percentage increases.

Based on Figure 5.4, we can observe a linear trend of the increase in reduction of carbon emissions when the percentage of renewable energy share increases, and the change in total carbon emission is shown in Figure 5.5. Our model is very sensitive to the change in renewable energy percentage in electricity generation, but we think this is reasonable since ways of generating electricity affect the carbon emission caused by HPC at the source.

5.7 Effects and Challenges of Switching to 100% Renewable Energy

As shown in Figure 5.5, the total HPC carbon emission falls sharply when all electricity is generated from renewable sources. After calculating, we see that HPC's total carbon emissions are hardly any. This is a good solution to the environmental problem of HPC's carbon emissions. We are interested in the benefits and challenges of achieving it, and summarize the **benefits** into two aspects: **Reduce Pollution and Salubrity**.

Reduce Pollution: According to Assumption 2, renewable energy sources like solar and hydropower produce little emissions. This shift reduces air pollution, mitigates the effects of climate change, and improves the overall environmental quality by lowering the concentration of pollutants in the air, soil, and water.

Salubrity: Reducing air pollution could increase air quality, which is great for public health. Fossil fuels cause a lot of pollution, leading to asthma, heart problems, and other illnesses. Renewable energy, on the other hand, doesn't pump out harmful emissions, which creates a cleaner and healthier environment for everyone.

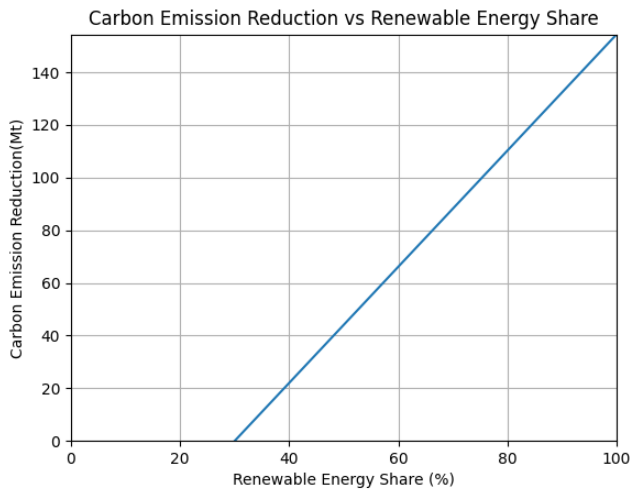


Figure 5.4: Reduction of Carbon Emission

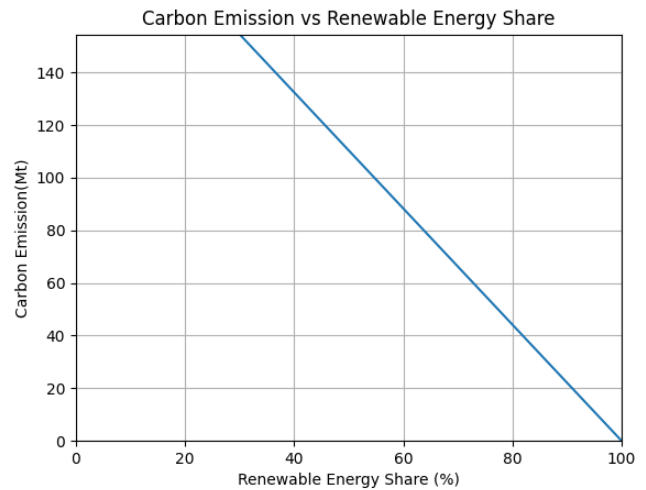


Figure 5.5: Change of Carbon Emission

However, we still have a long way to achieve the full application of renewable energy. These are some **challenges** of achieving it, and we summarize them into two aspects: **Variability and Social Impact**.

Variability: Unlike fossil fuels, renewable energy sources are intermittent[22], which means that they are not always available when demand is highest. This can lead to issues with energy reliability and grid stability, especially in regions where renewable resources are not consistently available throughout the year[23]. An unstable power supply can have serious consequences, such as the inability to maintain hospitals and fire operations. To overcome this challenge, solutions such as energy storage technologies (e.g., batteries), grid modernization, and the integration of diverse renewable sources are critical. Moreover, finding stable renewable energy sources is also important. For instance, building a Space base or Dyson sphere could maintain uninterrupted solar power[24].

Social Impacts: The initial investment required for new infrastructure—such as solar panels, wind turbines, and energy storage—is substantial. This can lead to deficit budgets, especially for regions with limited resources. Moreover, replacing fossil fuels is an industrial revolution that may be held back by many traditional energy companies. On the other hand, the transition could lead to job displacement in traditional energy sectors. Many workers in these industries might face challenges finding new jobs, which could create social and economic inequalities[25]. It's important to not only focus on the environmental benefits but also consider how to manage these transitions in a way that's fair to all communities.

6 Model of Estimating Future Trend

After estimating the carbon emissions of HPC systems in 2024, we shift our focus to examining the long-term trends in the HPC carbon footprint. To achieve this, we refine our model to analyze how carbon emissions change across the production, operational, and disassembly stages of HPC systems. According to [10], it is reasonable to conclude that the carbon production amount due to the production of HPC stays the same every year. Therefore, we will mainly focus on the operational and disassembly stages.

We investigate the trends in PUE, processor power, number of IT components, and carbon intensity. We also find that **upgrading GPU systems** emerges as a critical factor: while these upgrades reduce carbon emissions in the long run due to improved efficiency, they also cause an increase in emissions during the initial years.

6.1 Predicting Carbon Emission of Operational Stage

6.1.1 Trend in PUE

One significant factor that could influence the future energy consumption of HPC systems is **improved energy efficiency**, as represented by PUE values. Historically, PUE has shown a consistent downward trend, and it is expected to **continue declining in the future** due to advancements in technologies such as improved cooling systems and energy-proportional computing [26]. Assuming all other factors remain constant, a lower PUE will

result in reduced E_{IT} and a smaller carbon footprint. Therefore, the future PUE values of HPC should be estimated when considering carbon emissions during the operational stage. Therefore, we consider the historical data of PUE in data centers [26] [27] and try to predict future values.

Considering our small sample size and uncertainty in data trend, we decided to use **Grey Data Modeling** to model the PUE trend. Grey Data Modeling is used to model and predict the behavior of systems with limited or incomplete information. The term "grey" represents that the system lies between complete knowledge (white) and complete uncertainty (black). We employ the framework **GM(1,1)**, where "1,1" indicates a first-order differential equation with a single variable.

The key idea is to smooth the original data through an Accumulated Generation Operation (AGO), which reduces random fluctuations and reveals the underlying trend. This transformed data is then modeled using the Equation 6.1.

$$\frac{dx^{(1)}}{dt} + ax^{(1)}(t) = b \quad (6.1)$$

where a is the **development coefficient**, capturing the **internal characteristics of the system**, and b is the **grey action quantity** that represents external influences. The parameters a and b are estimated using the **least-squares method** on the cumulative data. Once the differential equation is established, its solution provides the **cumulative predictions**. The Grey Prediction Model is particularly suitable for predicting the PUE trend because it is highly efficient when working with **small datasets and systems with limited or uncertain data trends**. This ensures a relatively consistent and reliable prediction of future PUE trends, even when the available data is limited in quantity or exhibits variability. After utilizing Grey Data Modeling, the predictions are shown in Figure 6.1.

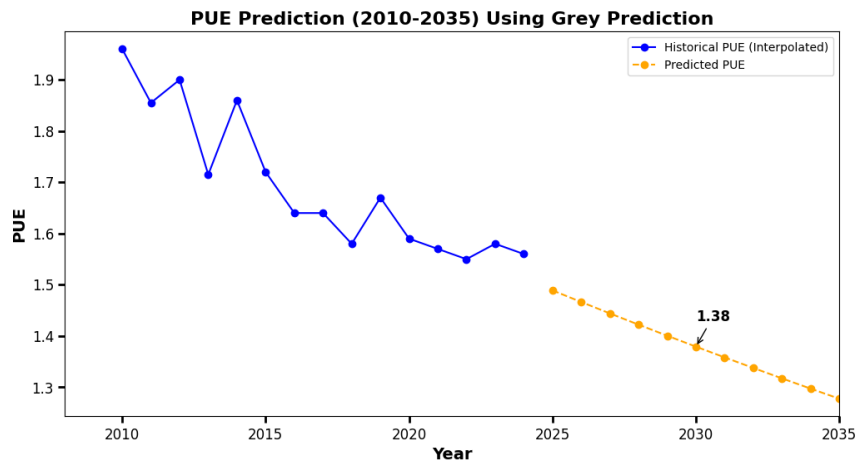


Figure 6.1: PUE trend

The prediction indicates a **continued gradual decline** in Power Usage Effectiveness (PUE) from 2025 to 2035, reaching a value of **1.379 by 2030**, as shown in Table 6.1. This trend is surprisingly consistent with the prediction in [29], which suggests a linear decline in PUE. This aligns with data centers focusing heavily on improving infrastructure efficiency to reduce operational costs. It is also driven by innovations such as liquid cooling that increases energy efficiency. Therefore, the trend should be reliable in the short run.

Year	2025	2026	2027	2028	2029	2030
PUE Prediction	1.489	1.466	1.444	1.422	1.401	1.379

Table 6.1: PUE Predictions for 2025-2030

6.1.2 Predicting Processor Power Growth

Our model calculates the electricity cost of HPC from basic IT components. We predict their rate of growth in power in the future. By analyzing the average power consumption of GPUs and CPUs from 2010 to 2024, we identify an annual growth rate of **14%**. However, the annual growth rate varies by 7%. In contrast, other components, such as storage and memory, show no significant trends in power consumption changes. Therefore, these components are excluded from our model to maintain focus on the most impactful factors.

6.1.3 HPC Market Growth and Increase in IT components

Since total energy consumption is the product of PUE and E_{IT} (Equation 4.3), the increasing use of IT components in HPC systems driven by applications such as AI will demand more energy, thereby generating higher carbon emissions. A key driver of this trend is the rapid growth of the HPC market. This growth directly translates to an increase in IT components such as CPUs and GPUs, which will raise E_{IT} .

After data collection of server and storage numbers from 2015 to 2024 and assuming a constant growth rate of GPUs and CPUs, we find out that the annual growth rate of storage is **6.16%** and the annual rate of sever representing the combination of processors, memory and network to be **5.91%**, according to Equation 4.5. Therefore, we can calculate the change in E_{IT} as Equation 6.2.

$$\Delta E_{IT} = \sum_{i \in \{IT \text{ components}\}} P_{average,i} \times \Delta N_i \times T \quad (6.2)$$

where ΔE_{IT} is the change in energy consumption of IT equipment, i is the different IT components, which include processor, storage, memory, and network. $P_{average,i}$ is the average power consumption of component c , ΔN_i is the change in the number of component i and T is the time, accounted as a whole year by Assumption 1.

Note that the logic here is to hold $P_{average}$ constant to emphasize the impact of the number of components (ΔN). The influence of $P_{average}$, which comprises P_{idle} , P_{max} , and utilization (U), will be discussed in the section on GPU upgrades, as these factors directly influence carbon emissions.

According to Equation 4.3, the total energy consumption of the facility is the product of PUE and E_{IT} , and the prediction for 2025 to 2030 is illustrated in Table 6.2.

Year	2025	2026	2027	2028	2029	2030
Predicted $E_{facility}$ (Billion kWh)	408.45	443.55	486.41	537.47	588.02	642.96

Table 6.2: Prediction of $E_{facility}$ from 2025 to 2030

6.1.4 Predicting Worldwide Carbon Intensity

Since carbon emission of renewable energy is nearly zero, we can simulate the trend of its proportion taking part in electricity generation to estimate the change of worldwide carbon intensity in the future. Figure 6.2 is the graph of the percentage of renewable, nonrenewable, and nuclear source electricity generation from 2000 to 2023:

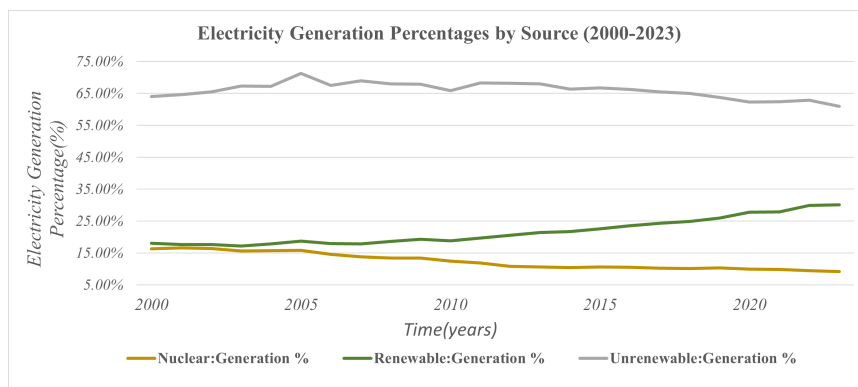


Figure 6.2: Electricity Generation Percentage

Based on the graph, we can observe a decreasing trend in nonrenewable energy percentage and an increasing trend in renewable energy percentage. Thus, it is plausible to speculate that there will be a decrease in carbon emissions in the near future due to the increasing use of renewable energy and less nonrenewable energy. We first try to fit various functions to the data, and this is shown by Figure 6.3.

Compared with linear and exponential fitting lines, the polynomial function is the best-fitting line with less error with actual data. Moreover, the function aligns with our prediction in the future. Although there is a global

push toward increasing renewable energy, factors such as limited infrastructure, intermittency, economic costs, and grid integration challenges still constrain the rapid acceleration of renewable adoption due to previous study [31]. Therefore, we deem the growth rate of renewable electricity to be linear, which validates a quadratic function. The following is the polynomial function after curve-fitting.

$$P(t) = 0.00492 \times t^2 - 19.46 \times t + 19406.9 \quad (6.3)$$

Given the prediction in electricity generation percentage, we model future CI by drawing a best-fitting curve of past CI data. Here, after using the standard Gaussian function and quadratic function, we finally decided to use the modified Gaussian function to be the best-fitting curve due to its small error. This is shown by Figure 6.4

Trend Fitting for Renewable Electricity Generation Percentage

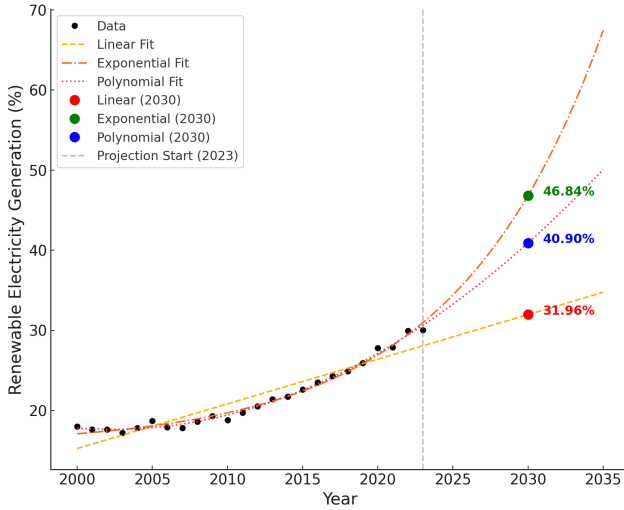


Figure 6.3: Electricity Generation % Trend¹

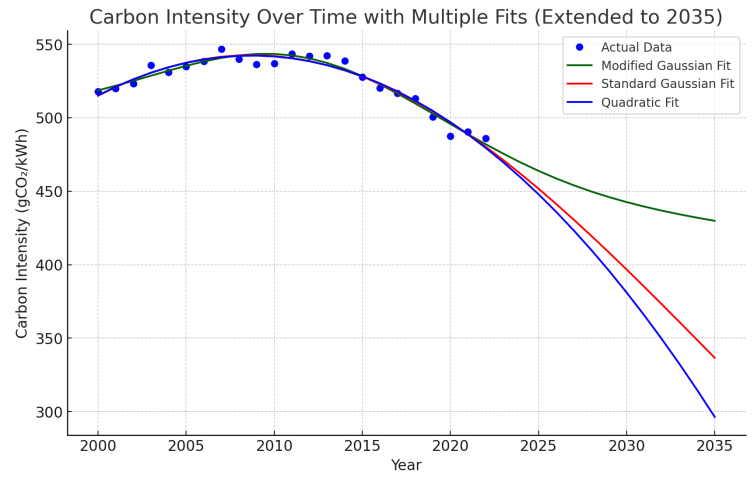


Figure 6.4: Fitting Curves of CI_{world} ²

$$Q(t) = 73.29 \times e^{-\frac{(t-2011.24)^2}{139.1112}} - 408.49 + \frac{877.24}{1 + 0.002 \times (t - 2011.24)} \quad (6.4)$$

Compared with the standard Gaussian function, the modified Gaussian function can better fit the overall trend in the past twenty years. This also follows what we expect will happen in the future. The technology and related efficiency of using energy will not always develop at a constant rate. Due to previous study[32], both of them are developing at a decreasing rate. Consequently, the carbon intensity will decrease at a decreasing rate. Ultimately, the tough situation may lead to a plateau in carbon intensity levels.

When taken together, the predicted decrease in electricity generation percentage trend and the predicted increase in future CI align with each other as the proportion of renewable electricity generation increases, less carbon is emitted per kWh of electricity generated, which leads to smaller CI . Finally, we use the best-fitting line of CI_{world} to simulate the value of worldwide carbon intensity in 2030.

6.1.5 Predicting Total Carbon Emission Operational Stage

According to the **Model of HPC Carbon Emission**, the total carbon emission due to operation E_{op} is the product of $E_{facility}$ and carbon intensity (CI). We have already predicted both $E_{facility}$ and CI. After multiplying them together, the results of C_{op} are shown in Table 6.3.

Year	2025	2026	2027	2028	2029	2030
C_{op} (ton CO ₂)	1.9554×10^8	2.1054×10^8	2.2695×10^8	2.4688×10^8	2.6741×10^8	2.8875×10^8

Table 6.3: Prediction of C_{op} from 2025 to 2030

¹Source: https://github.com/reformship/hpc_research/blob/main/EI-Stats-Review-All-Data.xlsx

²Source: https://github.com/reformship/hpc_research/blob/main/carbon-intensity-electricity.xlsx

6.2 GPU Upgrades and Increased Disassembly

As HPC systems evolve, GPU upgrades aim to enhance efficiency but may increase energy use and carbon emissions. High-end GPUs can raise data center energy consumption by up to 15% if not optimized [34]. Given their impact on **HPC carbon emissions**, further investigation is needed.

Questions

- (1) How do carbon emissions change over time with GPU upgrades, both in the short run and the long run?
- (2) Given the increase in replacements of GPUs, how will the processes of disassembly affect carbon emissions?

To facilitate the model of determining carbon changes due to upgrades, we deem the following assumptions to be logical and necessary.

Assumptions

- (1) A new generation of GPUs becomes available every 2 years. Based on historical data[35], we found the upgrade frequency of past GPUs is 2 years. Therefore, we assume that every 2 years, there will be a new generation of GPU.
- (2) Based on studies, GPUs have a maximum operational lifespan of 5 years, and they will be considered useless and be replaced after 5 years [36].
- (3) Upgrades between different GPU generations yield varying levels of carbon savings, with efficiency improving with each new iteration[10].
- (4) There is a higher probability of upgrading to the latest GPU version that fits within their budget. The newer the generation of GPU is, the more calculating ability it will have, and the more likely a data center will choose it.
- (5) The carbon emissions from disassembling GPUs remain constant over extended periods. The carbon emissions mainly come from plastics in GPU; there is not enough data to support the future percentage and weight of plastic in GPU. Therefore, we take it as a constant.

We define GPU generations using integers g , where **lower numbers represent older generations and higher numbers represent newer ones**. For example from a current standpoint, NVIDIA A100, V100 and H100 would be generations 0, 1 and 2. According to Assumption 5, it is reasonable to say that the number of generations will continue to grow. This framework allows us to track and simulate the replacement and upgrade processes for GPUs over time.

According to Assumption 4, we model the probability of GPU replacement increasing as the GPU approaches its maximum lifespan. We define the replacement probability as the following:

$$P(a) = \frac{a^2}{25} \quad (6.5)$$

where $P(a)$ is the replacement probability, and a represents the age of the GPU. This quadratic relationship reflects our observation that the likelihood of replacement grows non-linearly with age. As a approaches 5 (the maximum lifespan), $P(5) = 1$, meaning all GPUs of age five are replaced. We justify this form as following two reasons: GPUs closer to five years are increasingly likely to be replaced due to diminishing performance, and the energy consumption E per day remains relatively constant, so the probability of replacement grows uniformly with time.

Therefore, we can calculate the number of GPUs replaced for a specific generation g and age a using the equation below:

$$R_{g,a} = N_{g,a} \times \frac{a^2}{25} \quad (6.6)$$

Here, $N_{g,a}$ is the number of GPUs of age a in generation g . By multiplying $N_{g,a}$ with expectation values instead of real-time probabilities, we ensure stability in our simulation results. When a GPU is replaced, we model the probability of upgrading from an older generation g_1 to a newer generation g_2 according to the following:

$$P(g_2, g_1) = \frac{g_2 - g_1}{\sum G} \quad (6.7)$$

Here, $\sum G$ is the total difference between g_1 and all newer generations, defined in Equation 6.8.

$$\sum G = \sum_{g \in G_{\text{newer}}} (g - g_1) \quad (6.8)$$

where G_{newer} represents all generations newer than g_1 . Suppose $g_1 = 3$ and $g_2 = 5$. the newer generations are 4 and 5. Using Equation 6.7, the probability of upgrading from generation 3 to generation 5 is $\frac{5-3}{(5-3)+(4-3)} = \frac{2}{3}$. This approach ensures that **newer GPU generations are more likely to replace older GPUs**, as probabilities increase with the generational difference $g_2 - g_1$. This is consistent with Assumption 4 ("individuals have a higher probability of upgrading to the **latest** GPU version that fits within their budget"). Moreover, the likelihood of an upgrade depends on the total number of available newer generations and their differences from g_1 .

After introducing the basic variables of our simulation, we will introduce the full process of our simulation in one year in the following pseudo-code: This pseudo code shows the process of acting the upgrade of each GPU

Algorithm 1: Part 2: GPU Growth Simulation

```

1 GPU replacement:
2 Process Each Generation:
3 for each generation gen from 0 to newest_generation do
4   if gen  $\neq$  newest_generation then
5     for each age age from 0 to 5 do
6       for each newer generation newgen from gen + 1 to newest_generation do
7         GPU[newgen][0]  $\leftarrow$  GPU[newgen][0] + GPU[gen][age]  $\cdot$   $P(\text{newgen}, \text{gen}) \cdot R(\text{gen}, \text{age})$ 
8         GPU[gen][age]  $\leftarrow$  GPU[gen][age]  $\cdot$  (1 -  $R(\text{gen}, \text{age})$ )
9       end
10    end
11  else
12    GPU[gen][0]  $\leftarrow$  sum(GPU[gen])  $\cdot$  1.09
13  end
14 end
15 Age and Carbon Update:
16 for each generation gen from 0 to newest_generation do
17   for each age age from 0 to 2 do
18     Replaces[age]  $\leftarrow$  ReplaceNum(GPU[gen][age], age)
19     UpgradeCarbon  $\leftarrow$  UpgradeCarbon + SaveNum(Replaces[age], age)
20   end
21 end
22 if year % 2 = 0 then
23   newest_gen  $\leftarrow$  newest_gen + 1
24 end
25 year  $\leftarrow$  year + 1

```

group with different age and generation, and then calculate the increased number in each generation.

According to previous study [10], we observe that carbon savings from GPU upgrades vary depending on the GPU's age, which is consistent with the carbon savings above. The saved amount of carbon each year after one upgrade are: -100 in the first year, 75 in the second year and 100 in the third year with unit (g/kWh) For each year, we add up all the carbon changes brought by GPU upgrades as upgrade-saved carbon.

After calculation, the final carbon emissions **generated due to burning** and **saved because of GPU upgrade** are shown in Table 6.5. The burning is calculated in the same way in our previous model through

$$C_{\text{burn}} = N_{\text{burn}} \times C_B$$

with the discarded numbers coming from our simulation. C_B refers to the carbon emissions produced by burning a single GPU, which is 250g [18]. The following are the results of burn-caused carbon and upgraded reduced carbon with different units.

Table 6.4: Carbon Emissions Due to Burning and GPU Upgrades (Unit: Tons)

Year	Increased Emissions Due to Burning (Ton)	Reduction Due to GPU Upgrade (Ton)
2025	1166	2.85×10^7
2026	1459	4.76×10^7
2027	1198	7.628×10^7
2028	880	1.0795×10^8
2029	1185	1.3886×10^8
2030	1237	1.72×10^8

6.3 Results from 2025 to 2030

Adjusting Equation 5.1, we sum carbon emissions in all stages and then subtract the reduced carbon emission due to GPU upgrades to get the final result. Equation 6.9 shows the final calculation of carbon from 2025 to 2029.

$$C_{\text{total}} = C_{\text{prod}} + C_{\text{op}} + C_{\text{dis}} - C_{\text{up}} \quad (6.9)$$

C_{prod} represents the carbon emissions from the production of the equipment or system. C_{op} represents the operational carbon emissions, C_{dis} represents the carbon emissions from the disassembly of the equipment, and we minus C_{up} , which represents the reduced carbon emissions due to upgrading GPU. Figure 6.5 demonstrates the result of our prediction model.

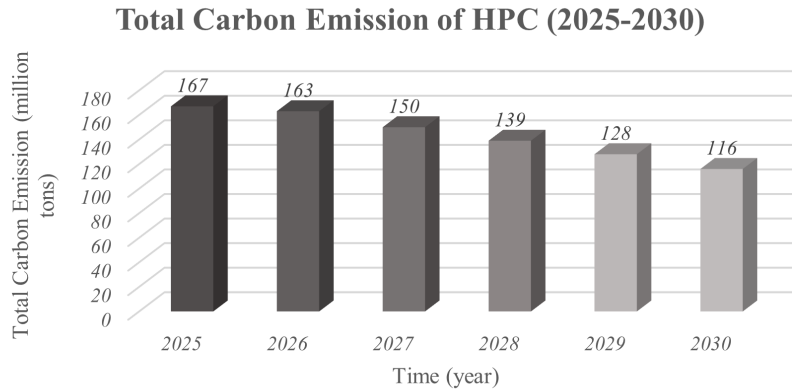


Figure 6.5: Predicted Carbon Emission of HPC Systems

From 2025 to 2030, the total carbon emissions of HPC show a **continuous and steady decrease**. The decline follows a roughly **linear trend**. This shows that while carbon emissions naturally increase due to the growing power and expansion of the HPC market, the effects are counteracted by **efficiency improvements** due to GPU upgrades, PUE improvement, and a decrease in carbon intensity. The decrease in PUE and CI values slows down the increasing carbon emissions. While the trend of C_{op} is increasing, the total carbon is decreasing. This illustrates the critical role of GPU upgrades. By prioritizing technological advancements in GPUs, HPC systems can maintain high performance while significantly reducing their environmental footprint.

To account for uncertainties in the model, we estimated the bounds of carbon emissions for 2030. Accounting for a potential 7% fluctuation in power, we incorporate an error margin of $\pm 5\%$ in our final solution for carbon emissions, accounting for other constant values such as the carbon emission during production. Finally, we determined the range of HPC carbon emissions in 2030 to be between **110.2 million tons and 121.8 million tons**.

6.4 Sensitivity analysis

To evaluate the robustness of our model, we conducted a sensitivity analysis on the proportion of processor energy within the total energy consumption of IT components. In our calculation of HPC carbon emissions, the energy consumption of processors (e.g., GPUs) plays a crucial role, as it is used to determine the total E_{IT} . The result is shown in Figure 6.6.

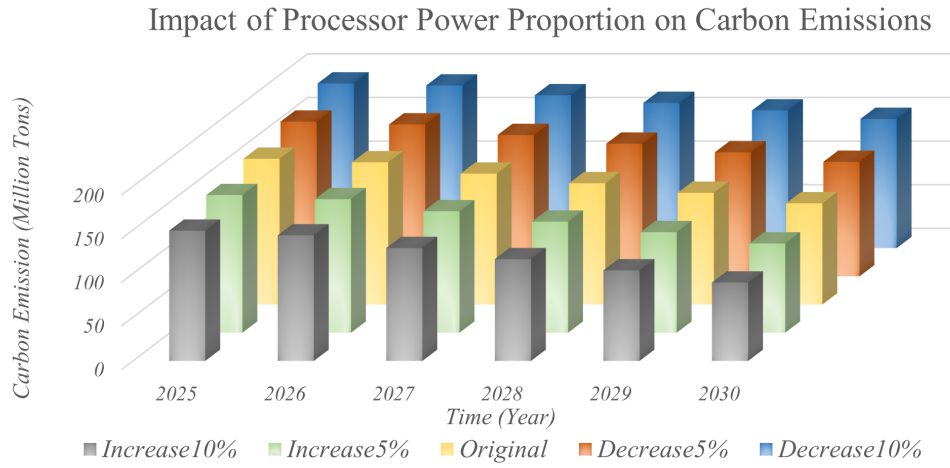


Figure 6.6: Sensitivity Analysis Result of GPU Power Percentage in Energy Consumption of IT Equipments

The sensitivity analysis shows a consistent trend: carbon emissions **steadily decrease over time**, regardless of changes in the processor power proportion. This observation aligns closely with our model's predictions for carbon emissions. Furthermore, the graph illustrates that a decrease in the processor power proportion leads to a roughly linear increase in total carbon emissions each year. This trend suggests that variations in the proportion of processor power have a **limited influence** on the total emissions. Therefore, our calculation method should give a comprehensive analysis of total carbon emissions.

7 Model of Impacts of Water Usage

Then, we turn our focus from carbon emissions to water pollution and consumption of HPC. Our model includes all water consumption and pollution of the process from electricity generation to cooling system of HPC systems. It classifies the cooling system into two types and calculates them separately.

7.1 Why This Aspect?

According to a previous study [37], despite carbon emission, significant water consumption is a corollary of HPC systems to work at high efficiency. Among several usages, cooling systems contribute a large portion of water consumption, discharging water vapor, a kind of greenhouse gas [38]. Thus, it is necessary to consider water consumption and potential pollution used by HPC systems.

7.2 Water Consumption of Electricity Generation

We first find out the average water consumption of electricity generation using different technologies.

Technology	Coal/Oil	Nuclear	Gas	Hydro	Solar PV	Wind	Biomass
Water Consumption rate (L/kWh)	2.4	2.5	0.9	70	0.4	0.3	90

Table 7.1: Average Water Consumption of Electricity Generation of Various Technology

Furthermore, we find out the total electricity consumed by HPC systems. According to *Assumption 3*, we can calculate the respective proportion of each technology in total generation and times them to the corresponding water consumption rate, and then calculate the sum of the results, which is the total water consumption of electricity consumed by HPC systems **in electricity generation stage**.

$$W_{EG} = \sum_{i=1}^P WR_i \times \frac{EG_i}{EG_{total}} \times E_{HPC}$$

while W_{EG} stands for water consumption of electricity generation, and the WR_i stands for water consumption rate of fuel type i .

7.3 Water Consumption of HPC Systems

Most of the water consumption of HPC systems is used by cooling systems, which can be divided into three separate parts: direct liquid cooling, air cooling, and phase-changing cooling.

Direct liquid cooling passes cooling liquids near the processors to cool them down. Around every processor, there is a pipe to get water in touch with them to take away their heat. According to the studies we've found[39], the water consumption of this type of cooling system is

$$W_{Liq} = WR \times N_p \times T$$

where WR is the average flow rate in the pipes near the processors, N_p is the number of processors, and T is hours in a year, which is $365 \times 24h$.

Unlike direct liquid cooling, **air cooling** primarily uses air as a medium of heat dissipation. According to a previous study, the air cooling system can be fundamentally divided into three types of cooling systems [40]: Dry Air Cooler (DAC), Spray Type Air Cooler (SAC), and Evaporative Air Cooler (EAC).

For dry air cooler (DAC), $W_{DAC} = 0$, which means there is **negligible water consumption** in this system.

For spray air cooler (SAC), the water consumption can be calculated with Equation 7.1:

$$W_{SAC} = 1.3 \times W_A \times \frac{X_I - X_S}{\rho_s} \quad (7.1)$$

where X_I stands for the absolute humidity of outlet air, and X_S stands for the absolute humidity of inlet air. ρ_s stands for the density of spray water.

For evaporative air cooler (EAC), the water consumed is **almost the same as SAC**. The calculation can use the identical equation.

Thus, the total water consumption can be calculated with Equation 7.2:

$$W_{air} = W_{DAC} + W_{SAC} + W_{EAC} \quad (7.2)$$

For phase-changing cooling, indirect liquid cooling takes most of the water consumption. Indirect cooling is the process of transferring heat indirectly via a cooling medium (usually a liquid or air) rather than directly contacting the heat source. This approach is widely used in data centers, industrial cooling, and HVAC systems.

Phase-changing cooling (PCC): PCC is a cooling technology that uses the process of phase change to move heat more efficiently. When the liquid evaporates, it absorbs much energy to change the phase from liquid to gas, decreasing the surrounding temperature within the HPC system. PCC systems are usually designed to recycle the cooling water, which helps reduce water waste and makes them more sustainable. The water consumption is small, only consisting of **water leakage** and **blow down**[41].

$$W_c = W_l + W_b \quad (7.3)$$

In this equation, W_c is the total water consumption in this system, W_l is the water waste due to leakage, and W_b is the water waste due to blowing down. Then, we need to calculate W_l and W_b separately. The proportion of water usage is mentioned in [42].

As we mentioned above, water evaporation does not cause consumption due to the recycling design, but we can use it to calculate leakage (drift in Figure 8.1) and blow-down. Because we want to focus on the total water consumption of the HPC cooling system, the water pollution of the occupants is not taken into account. W_e , which stands for water evaporating in the cooling system, can be calculated as below.

$$W_{Eva} = \frac{Q}{L} \quad (7.4)$$

In this equation, Q stands for the heat energy removed by evaporation, and L stands for the latent heat of vaporization for water.

$$Q = k \times E_c \quad (7.5)$$

Annual Water Consumption for a Data Center

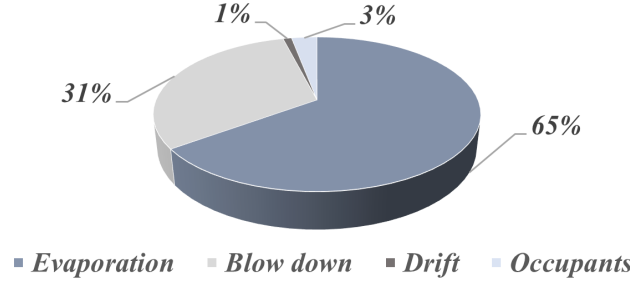


Figure 7.1: Proportion of water usage in a data center

In this equation, k is a constant that represents the percentage of electricity that is converted to heat[34], and E_c is the electricity consumed by all HPCs in global that use a phase-changing cooling system for one year. This emphasizes the relationship between energy consumption and heat, as well as water consumption.

According to these equations, we could obtain the final water consumption of the phase-changing cooling system in a year using the proportion of evaporation, blow down, and drift.

7.3.1 Total Water Consumption

Considering three separate cooling systems, we can find the total accumulation of water consumption of the cooling system in general.

$$W_{cool} = W_{Liq} + W_{Air} + W_{Eva}$$

Ultimately, the sum of water consumption of both the electricity generation stage and cooling system is the final result of our model, total water consumption.

$$W_{tot} = W_{cool} + W_{EG}$$

7.4 Water Pollution of HPC Systems

To calculate the amount of pollutants discharged by HPC systems, we need to quantify the pollutant loads from various sources. The total mass of pollutants, denoted as L , is the sum of the masses of individual pollutants from different sources:

$$L = L_{cooling} + L_{electricity} \quad (7.6)$$

where $L_{cooling}$ is the total mass of pollutants discharged from the cooling systems, and $L_{electricity}$ is the total mass of pollutants associated with electricity generation for HPC systems.

Pollutant Load from Cooling Systems: The mass of pollutants discharged from cooling systems can be calculated using:

$$L_{cooling} = \sum_i (C_i \times V_{discharge}) \quad (7.7)$$

where C_i is the concentration of pollutant i in the discharged water, $V_{discharge}$ is the total volume of water discharged from the cooling system per year; the summation is over all pollutants i present in the discharge.

Pollutant Load from Electricity Generation: The mass of pollutants associated with electricity generation can be calculated using:

$$L_{electricity} = E_{total} \times E_{pollutant} \quad (7.8)$$

where E_{total} is the total electricity consumption of HPC systems per year, $E_{pollutant}$ is the emission factor for the pollutant per unit of electricity generated.

7.5 Water Consumption and Pollution of HPC Systems in the USA

To better illustrate our model and how it leads to the final results, we use a real-life example of the USA to demonstrate how our calculations work.

Water Consumption: To determine the water consumption of HPC systems in the USA, we first identify all the variables required for each type of cooling system and for electricity generation. After calculating, we obtain the water consumption for each sector, as well as the total water consumption. The W_{tot} is equal to 33547 billion liters, according to Table 7.2.

Water Consumption	Air Cooling	Phase Change	Liquid Cooling	Electricity Generation
Billion Liters	507.06	37.56	81.83	32921

Table 7.2: Water consumption of each type

Water Pollution: Similarly, we calculate the water pollution of HPC systems in the USA by determining the mass of pollutants discharged from cooling systems and electricity generation. The total mass of pollutants discharged, L , is equal to approximately 5.15 million kilograms per year.

7.6 How It Relates to Other Aspects?

While calculating water consumption, it is important to recognize its interplay with other critical factors, such as energy consumption. Our model accounts for water consumption across the entire life cycle, including electricity generation and cooling systems, demonstrating a direct correlation between energy and water usage. For instance, changes in cooling systems can influence the Power Usage Effectiveness (PUE) of HPC systems, thereby affecting both energy and water consumption.

Moreover, water usage introduces environmental risks, including potential water pollution. During processes such as dilution, improper management of cooling systems may lead to chemical leakage, posing a threat to the natural environment.

8 Recommendations to Reduce Environmental Impact of HPC

8.1 Technical Solutions

Upgrading Power Stations: Improving this efficiency not only reduces the amount of fuel required for the same energy output but also decreases waste heat, minimizing the environmental impact.

GPU Upgrades: Through upgrading, GPUs' flop per energy is higher, leading to lower energy use of HPCs.

Automatic Energy Distribution: Take automatic energy distribution into HPC to monitor the power required for each part as HPCs work, making energy efficiency higher and lowering HPC energy consumption.

Improve Cooling System: Improving the conduits and power of water transferring, lowering energy consumption by cooling systems, which increases PUE value.

8.2 Policy-Oriented Solutions

Subsidies on clean energies: Supporting the development of clean energies, lowering the carbon intensity of the world, leading to less carbon emission in HPC.

WUE Lines: WUE (Water Use Efficiency) measures the water use efficiency for a data center, which could be calculated as follows.

$$WUE = \frac{W_{tot}}{E_{facility}}$$

Government could set a maximum value for WUE; if any HPC center's WUE goes beyond this line, they will be fined. In this way, water consumption is reduced because data centers would be trying to avoid heavy fines.

8.3 Incorporate the Recommendation to Our Model

After restricting the WUE value, the data centers would decrease their water usage by changing their cooling system to be more water-efficient. In previous calculations, we discovered that air cooling wastes more water and has a greater WUE. In order to achieve the purpose of reducing water consumption, they could change their cooling system from air cooling, which costs more water since there is no recycling design, to direct liquid cooling or phase-changing cooling. They may also improve the energy efficiency of their total system. By impelling them to do these efforts, the total water consumption would decrease after this policy is enacted.

9 Strengths and Weaknesses

9.1 Strengths

1. A full life-cycle perspective is employed, encompassing not only the operational stage (working and running phases) but also the production and disassembly stages. This holistic approach allows for a thorough evaluation of carbon emissions and carbon intensity across all stages of the HPC system life cycle.

2. By adopting a global perspective, our approach spans a wide range of applications rather than limiting the scope to specific types of HPC systems like those in the *TOP500* and *Green500*. This approach makes *Model of HPC Energy Consumption* more complete.

3. In *Model of Estimating Future Trend*, our method combines theoretical foundations with empirical data, ensuring both scientific rigor and practical applicability. For example, on the theoretical side, we calculate energy consumption starting from the component level, focusing on CPUs, GPUs, and other fundamental elements of HPC. This bottom-up approach improves model precision and reduces potential errors. On the empirical side, we incorporate relevant data to validate and support model results, further enhancing their reliability and applicability.

9.2 Weaknesses

1. Our models are subject to some limitations inherent to Life Cycle Assessment (LCA). For example, differences in the life cycle scope can lead to significant discrepancies. In our model, carbon emissions from combustion are attributed to the disassembly stage, while other studies may assign these emissions to disposal. Such variations reduce the comparability of results across cases.

2. Some prediction methods restrict the model from short-term predictions. For instance, the near-linear by Grey Data Modeling does not account for the **diminishing returns** on efficiency investments in the data center industry. Furthermore, if the current trend continues indefinitely, PUE would fall below 1.0, which is theoretically impossible. Therefore, though a nearly linear fit by the Grey Data Modeling in the **short run (2025 to 2030)** is effective, it will no longer be applicable in the long run.

3. Revolutionary technological advancements are not considered in the model. This exclusion may overlook significant future shifts in energy consumption or carbon emissions.

10 Discussion

Despite the depth of our analysis, there are opportunities to enhance and extend our models. Future research could focus on development models that cooperate with the emergence of new technologies. An example of expansion is to incorporate **neuromorphic processors** into our carbon emission models. Neuromorphic processors are energy efficient because they process information using spiking neural networks rather than traditional binary logic. This can lead to substantial reductions in power consumption and heat generation compared to conventional Processors. Adding neuromorphic processors to our models would change parameters related to processor power, utilization rates, and PUE. It would significantly alter the energy consumption and carbon emissions associated with HPC systems. Other technologies, such as quantum computing, can also be appended to our models. Adding these technological advances to our models can make our models more effective in predicting environmental impacts in the future.

References

- [1] Jones, N. (2018). How to stop data centres from gobbling up the world's electricity. *Nature*, 561(7722), 163–166. <https://doi.org/10.1038/d41586-018-06610-y>
- [2] O'Brien, K. D., Pietri, I., Ravi Shankar Reddy, Alexey Lastovetsky, & Sakellariou, R. (2017). A Survey of Power and Energy Predictive Models in HPC Systems and Applications. 50(3), 1–38. <https://doi.org/10.1145/3078811>
- [3] Uptime Institute. (2024). 14th annual global data center survey results 2024. Uptime Institute. <https://intelligence.uptimeinstitute.com/resource/uptime-institute-global-data-center-survey-2024>
- [4] Sterling, T., & Anderson, M. (2017). High performance computing: Modern systems and practices. CRC Press.
- [5] Lee, H. H., Park, J., & Kim, H. S. (2011). Energy efficiency in high performance computing systems. *IEEE Transactions on Parallel and Distributed Systems*, 22(2), 267–277. <https://doi.org/10.1109/TPDS.2010.229>
- [6] Hertwich, E. G., Gibon, T., Bouman, E. A., Arvesen, A., Suh, S., Heath, G. A., Bergesen, J. D., Ramirez, A., Vega, M. I., & Shi, L. (2014). Integrated life-cycle assessment of electricity-supply scenarios confirms global environmental benefit of low-carbon technologies. *Proceedings of the National Academy of Sciences*, 112(20), 6277–6282. <https://doi.org/10.1073/pnas.1312753111>
- [7] Dayarathna, M., Wen, Y., & Fan, R. (2016). Data Center Energy Consumption Modeling: A Survey. *IEEE Communications Surveys & Tutorials*, 18(1), 732–794. <https://doi.org/10.1109/comst.2015.2481183>
- [8] Shehabi, A., Smith, S., Sartor, D., Brown, R., Herrlin, M., Koomey, J., Masanet, E., Horner, N., Azevedo, I., & Lintner, W. (2016). United States Data Center Energy Usage Report. https://www.iea-4e.org/wp-content/uploads/publications/2016/06/05j_-_LBNL_-_US_Data_Centres_Energy_USe.pdf
- [9] Full-System Power Analysis and Modeling for Server Environments. (2006). <https://rivoire.cs.sonoma.edu/research/pubs/economou.2006.mantis.mobs.slides.pdf>
- [10] Li, B., Roy, R., Wang, D., Samsi, S., Gadepally, V., & Tiwari, D. (n.d.). Toward Sustainable HPC: Carbon Footprint Estimation and Environmental Implications of HPC Systems. Retrieved November 11, 2024, from <https://arxiv.org/pdf/2306.13177>
- [11] Colorlib. (n.d.). SEMI Report: Global Semiconductor Production Capacity Expected to Reach a Record 30 Million Wafers per Month in 2024 - SEMI Large Semiconductor Industry Network <https://www.semi.org.cn/site/semi/article/a60f27a777374e5999f55db4dfeacc52.html>
- [12] <https://hyperionresearch.com/wp-content/uploads/2021/09/Hyperion-Research-HPC-MCS-Storage-File-System-Interconnects-September-2021.pdf>
- [13] National Grid. (2024, March 27). What is nuclear energy? | Why is nuclear considered a clean energy? | National Grid Group. www.nationalgrid.com; National Grid. <https://www.nationalgrid.com/stories/energy-explained/what-nuclear-energy-and-why-it-considered-clean-energy>
- [14] Bouckaert, S., Pales, A. F., McGlade, C., Remme, U., Wanner, B., Varro, L., D'Ambrosio, D., & Spencer, T. (2021). Net Zero by 2050: A Roadmap for the Global Energy Sector. trid.trb.org. <https://trid.trb.org/view/1856381>
- [15] UNFCCC. (2024). The Paris Agreement. United Nations Climate Change; United Nations. <https://unfccc.int/process-and-meetings/the-paris-agreement>
- [16] <https://ourworldindata.org/grapher/carbon-intensity-electricity>
- [17] Boone, R. (2023, August 3). How much do your PC and components weigh? – everything you need to know. Ecsi. <https://ecsi.org/how-much-do-your-pc-and-components-weigh/#:~:text=Graphics%20Card%20%28GPU%29%3A%20Depending%20on%20the%20type%20and,solutions%20tend%20to%20be%20on%20the%20heavier%20side>
- [18] Alter, L. (2021, December 8). The carbon footprint of plastic is way higher than we thought. www.treehugger.com/carbon-footprint-of-plastic-way-higher-than-thought-5212191
- [19] <https://www.energyinst.org/statistical-review>
- [20] Norton, A. (n.d.). HPC and AI Processors: An Update May 2021. Retrieved November 14, 2024, from <https://www.hpcuserforum.com/wp-content/uploads/2021/05/Hyperion-Resarch-HPC-UF-Processor-Presentation-Norton-04.20.2021-1-1.pdf>
- [21] International Energy Agency. (2024, January). Electricity 2024 – Analysis. IEA. <https://www.iea.org/reports/electricity-2024>
- [22] Sovacool, B. K., et al. (2016). "The challenges of 100% renewable electricity." *Nature Communications*, 7, 12798.
- [23] Naylor, T. R. S., Kartha, S., & Lucas, M. (2020). Global pathways to 100% renewable electricity. *Nature Communications*, 11, 4202.
- [24] Dyson, F. (1960). Search for Artificial Stellar Sources of Infrared Radiation. *Science*, 131(3414), 1667-1672.
- [25] Healy, N., & Barry, J. (2017). "The economic and social implications of a transition to renewable energy." *Energy Policy*, 101, 176-183.
- [26] Luca Castellazzi, Maria, A., & Paolo, B. (2017, December 11). Trends in data centre energy consumption under the European Code of Conduct for data centre energy efficiency. <https://doi.org/10.2760/358256>
- [27] Liu, Y., Wei, X., Xiao, J., Liu, Z., Xu, Y., & Tian, Y. (2020). Energy consumption and emission mitigation prediction based on data center traffic and PUE for global data centers. *Global Energy Interconnection*, 3(3), 272–282. <https://doi.org/10.1016/j.gloe.2020.07.008>
- [28] Huang, X., Yan, J., Zhou, X., Wu, Y., & Hu, S. (2023). Cooling Technologies for Internet Data Center in China: Principle, Energy Efficiency, and Applications. *Energies*, 16(20), 7158. <https://doi.org/10.3390/en16207158>
- [29] Avelar, V., Electric, S., Azevedo, D., French, A., & Network Power, E. (2012). PUETM: A COMPREHENSIVE EXAMINATION OF THE METRIC EDITORS. https://datacenters.lbl.gov/sites/default/files/WP49-PUE%20A%20Comprehensive%20Examination%20of%20the%20Metric_v6.pdf <https://www.google.com/about/datacenters/efficiency/>
- [30] Worldwide HPC Market Forecast by Vertical, 2020-2025 - Hyperion Research. (2021, November 15). Hyperion Research. <https://hyperionresearch.com/product/worldwide-hpc-market-forecast-by-vertical-2020-2025/>
- [31] Salem, M., Kamarol, M., Das, H. S., Ang, T. Z., & Nazari, M. A. (2022). A comprehensive study of renewable energy sources: Classifications, challenges, and suggestions. *Energy Strategy Reviews*, 40, 100813. <https://doi.org/10.1016/j.esr.2022.100813>
- [32] Chen, B., Ciais, P., & Wang, T. (2022). Challenges to achieve carbon neutrality of China by 2060: Status and perspectives. *Science Bulletin*. <https://doi.org/10.1016/j.scib.2022.06.013>
- [33] Energy Efficiency in High-Performance Computing: Balancing Speed and Sustainability. (2023, November 14). NVIDIA Technical Blog.
- [34] Zhang, Q., Zhani, M. F., Hellerstein, J. L., Yang, C., & Boutaba, R. (2016). Data center power consumption modeling and prediction through machine learning. *Proceedings of the 9th IEEE International Conference on Cloud Computing (CLOUD)*, 681–688.
- [35] <https://finance.sina.com.cn/jjxw/2024-06-04/doc-inaxpzzxi8063656.shtml>
- [36] HPCwire. (2021, January 4). What's new in HPC research: GPU lifetimes, the Square Kilometre Array, support tickets & more. HPCwire.
- [37] Silva, C. A., Vilaça, R., Pereira, A., & Bessa, R. J. (2024). A review on the decarbonization of high-performance computing centers. *Renewable and Sustainable Energy Reviews*, 189, 114019.
- [38] Held, I. M., & Soden, B. J. (2000). Water vapor feedback and global warming. *Annual review of energy and the environment*, 25(1), 441-475.
- [39] Daoist Baba. (2016, May 20). Analysis of cooling system for high-speed water-cooled permanent magnet motors <https://www.doc88.com/p-3019721990499.html>
- [40] Zhang, H., Feng, X., & Wang, Y. (2018). Comparison and evaluation of air cooling and water cooling in resource consumption and economic performance. *Energy*, 154, 157-167.
- [41] Yuan, X., Zhou, X., Pan, Y., Kosonen, R., Cai, H., Gao, Y., & Wang, Y. (2021). Phase change cooling in data centers: A review. *Energy & Buildings*, 236, 110764.
- [42] CIO. (n.d.). The convergence of high-performance compute and sustainability. CIO. <https://www.cio.com/article/306135/the-convergence-of-high-performance-compute-and-sustainability.html>
- [43] Raptis, C. E., Boucher, J. M., & Pfister, S. (2017). Assessing the environmental impacts of freshwater thermal pollution from global power generation in LCA. *Science of the Total Environment*, 580, 1014-1026.

11 Report on Use of AI

OpenAI *ChatGPT* (May 13, 2024 version, ChatGPT-4o)

Query1:<Curve fitting these data:Year Renewable Generation.

Year	2000	2001	2002	2003	2004	2005
Renewable:Generation (%)	18.00	17.60	17.60	17.20	17.80	18.70
Year	2006	2007	2008	2009	2010	2011
Renewable:Generation (%)	17.90	17.80	18.60	19.30	18.80	19.70
Year	2012	2013	2014	2015	2016	2017
Renewable:Generation (%)	20.50	21.40	21.70	22.60	23.50	24.30
Year	2018	2019	2020	2021	2022	2023
Renewable:Generation (%)	24.90	25.90	27.80	27.87	29.91	30.04

Give me linear, exponential, and polynomial functions. The graph should fit the 2035 year and include data for the 2030 year for each function.>

Output :<

```
# Plot data and future projections with updated title
and emphasized 2030 projections
plt.figure(figsize=(14, 6))
plt.plot(years, renewable_gen, 'o', label='Data', color='black')
plt.plot(x_future, linear_future, label='Linear Fit', linestyle='--')
plt.plot(x_future, exp_future, label='Exponential Fit', linestyle='-.')
plt.plot(x_future, poly_future, label='Polynomial Fit', linestyle=':')

# Marking the current data range and future projection
plt.axvline(x=2023, color='gray', linestyle='--', alpha=0.5,
label='Projection Start (2023)')

# Emphasize and annotate 2030 predictions for each model with bold text
plt.annotate(f'{linear_2030:.2f}%', (year_2030, linear_2030),
textcoords="offset points", xytext=(5, -10),
ha='center', color='orange', fontweight='bold')
plt.annotate(f'{exp_2030:.2f}%', (year_2030, exp_2030),
textcoords="offset points", xytext=(5, -10),
ha='center', color='red', fontweight='bold')
plt.annotate(f'{poly_2030:.2f}%', (year_2030, poly_2030),
textcoords="offset points", xytext=(5, -10),
ha='center', color='purple', fontweight='bold')

# Plot vertical line at 2030 for clarity
plt.axvline(x=2030, color='gray', linestyle='--', alpha=0.5,
label='2030 Projection')

# Add labels and updated title
plt.xlabel('Year')
plt.ylabel('Renewable Electricity Generation (%)')
plt.title('Trend Fitting for Renewable Electricity Generation
Percentage (Projection to 2035)')
plt.legend()
plt.grid(True)
plt.show()
```

This plot shows the projection of the renewable generation percentage trend up to the year 2035 based on the three curve fit models: linear, exponential, and polynomial. The dotted vertical line at 2023 marks the start of the

projection. Each model suggests different growth rates beyond 2023, with the exponential fit indicating a rapid increase, the polynomial fit showing moderate acceleration, and the linear fit continuing at a steady rate. Let me know if you need any further analysis on these projections! >

Query2:<

Year	2000	2001	2002	2003	2004	2005
Carbon Intensity (gCO ₂ /kWh)	517.84	520.08	523.33	535.74	530.92	534.97
Year	2006	2007	2008	2009	2010	2011
Carbon Intensity (gCO ₂ /kWh)	538.36	546.70	540.02	536.39	536.86	543.62
Year	2012	2013	2014	2015	2016	2017
Carbon Intensity (gCO ₂ /kWh)	542.00	542.37	538.84	527.64	520.15	516.63
Year	2018	2019	2020	2021	2022	
Carbon Intensity (gCO ₂ /kWh)	512.97	500.50	487.36	490.35	485.99	

Try to fit it with a Gaussian function, a Gaussian function with an asymptotic line that restricts the y variable to decrease sharply in the future, and a polynomial function, show the curve to 2035 year and show the data of 2030. >

Output :<

```
# Fit the standard Gaussian function to the data
standard_gaussian_params, _ = curve_fit(standard_gaussian_fit, years,
carbon_intensity, p0=[230, 2008, 17])

# Generate fitted values for the standard Gaussian function extended to 2035
standard_gaussian_fitted_values_2035 =
standard_gaussian_fit(extended_years_2035, *standard_gaussian_params)

# Plot data and fitted curves (modified Gaussian, standard Gaussian, and
quadratic) to 2035
plt.figure(figsize=(10, 6))
plt.plot(years, carbon_intensity, 'bo', label="Actual Data")
plt.plot(extended_years_2035, modified_fitted_values_2035, 'darkgreen',
label="Modified Gaussian Fit")
plt.plot(extended_years_2035, standard_gaussian_fitted_values_2035, 'red',
label="Standard Gaussian Fit")
plt.plot(extended_years_2035, quadratic_fitted_values_2035, 'blue',
label="Quadratic Fit")
plt.xlabel("Year")
plt.ylabel("Carbon Intensity (gCO\textsubscript{2} /kWh)")
plt.title("Carbon Intensity Over Time with Multiple Fits (Extended to 2035)")
plt.legend()
plt.show()
```

The plot includes the modified Gaussian, standard Gaussian, and quadratic fits, each extended to 2035. This allows a comparison of how the standard Gaussian differs from the modified version, especially in how it approaches zero more quickly without the asymptotic leveling-off effect. Let me know if further adjustments or explanations are needed!>

Query3:< Based on the data points, do a curve fit of logarithmic function and exponential function to 2030.>

Output :<

Year	2010	2011	2012	2013	2014
PUE	1.96	1.855	1.90	1.715	1.86
Year	2015	2016	2017	2018	2019
PUE	1.72	1.64	1.58	1.58	1.67
Year	2020	2021	2022	2023	2024
PUE	1.59	1.57	1.55	1.58	1.56

```
# Plot the results with integer years on the horizontal axis
plt.figure(figsize=(10, 6))
plt.scatter(x_regression, y_regression, color='blue', label='Actual Data')
plt.plot(years_extended, logarithmic_predictions, label='Logarithmic Fit (to
2030)', linestyle='--', color='purple')
plt.plot(years_extended, exponential_predictions, label='Exponential Decay
Fit (to 2030)', linestyle='--', color='brown')
plt.axvline(x=2024, color='red', linestyle=':', label='Prediction Boundary')

plt.title('Logarithmic and Exponential Decay Fits for PUE to 2030')
plt.xlabel('Year')
plt.ylabel('PUE')
plt.xticks(ticks=np.arange(2010, 2031, 1)) # Set horizontal axis to integer years
plt.legend()
plt.grid(True)
plt.show()
```

The plot now includes:

Logarithmic Fit (purple dashed line): Models the decrease with a diminishing rate, respecting the trend and keeping values above 1.0. Exponential Decay Fit (brown dashed line): Models the trend as a gradual approach to a limit (e.g., PUE nearing 1.0 but not crossing it). Both fits extend to 2030, with predictions starting after 2024 (indicated by the red line). Let me know if you need further refinements or insights! >